

MARKETPLACE APP

BICT2406 Mini Project – Group 3

Group Members

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1. Introduction

The Marketplace App is a mobile e-commerce application developed to provide users with a convenient platform for browsing products, viewing product information, adding products to a shopping cart, checking out, and making payments. The project also includes a web-based administration portal for managing marketplace products through Firebase.

2. Development Tools and Technologies

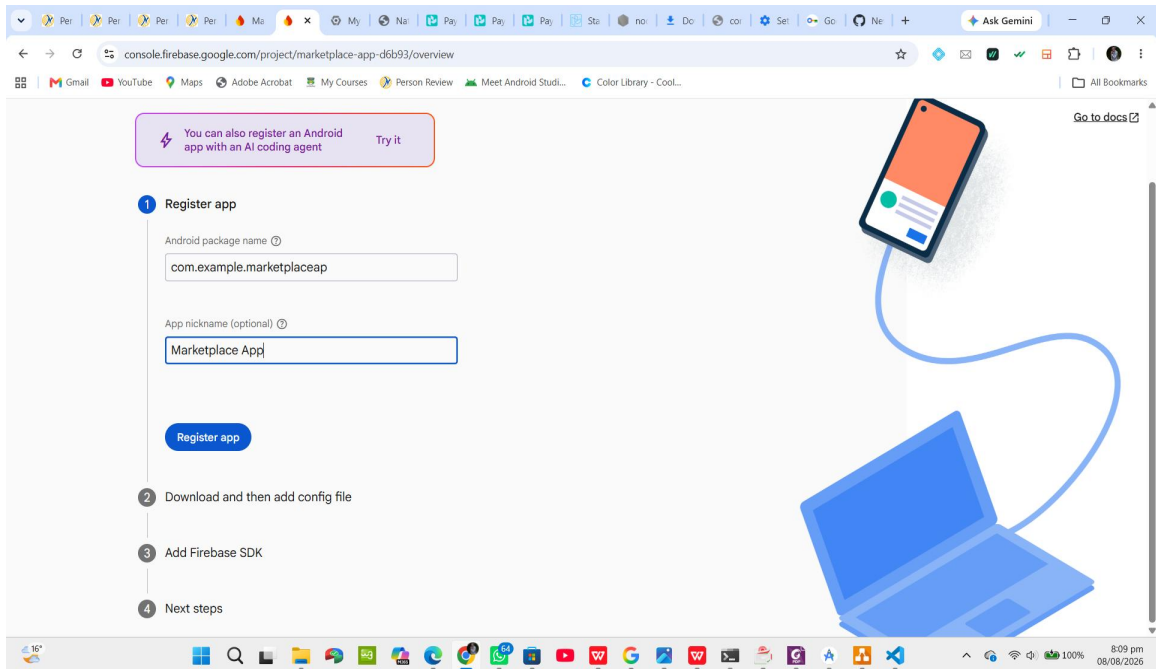
- Android Studio
- Kotlin
- Jetpack Compose
- Material 3
- Firebase Authentication
- Firebase Firestore
- Firebase Cloud Functions
- Firebase Hosting
- PayChangu
- React
- Vite
- GitHub

3. Project Planning and Android Studio Setup

The project was created in Android Studio using Kotlin and Jetpack Compose. The application package was configured as `com.example.marketplaceapp`.

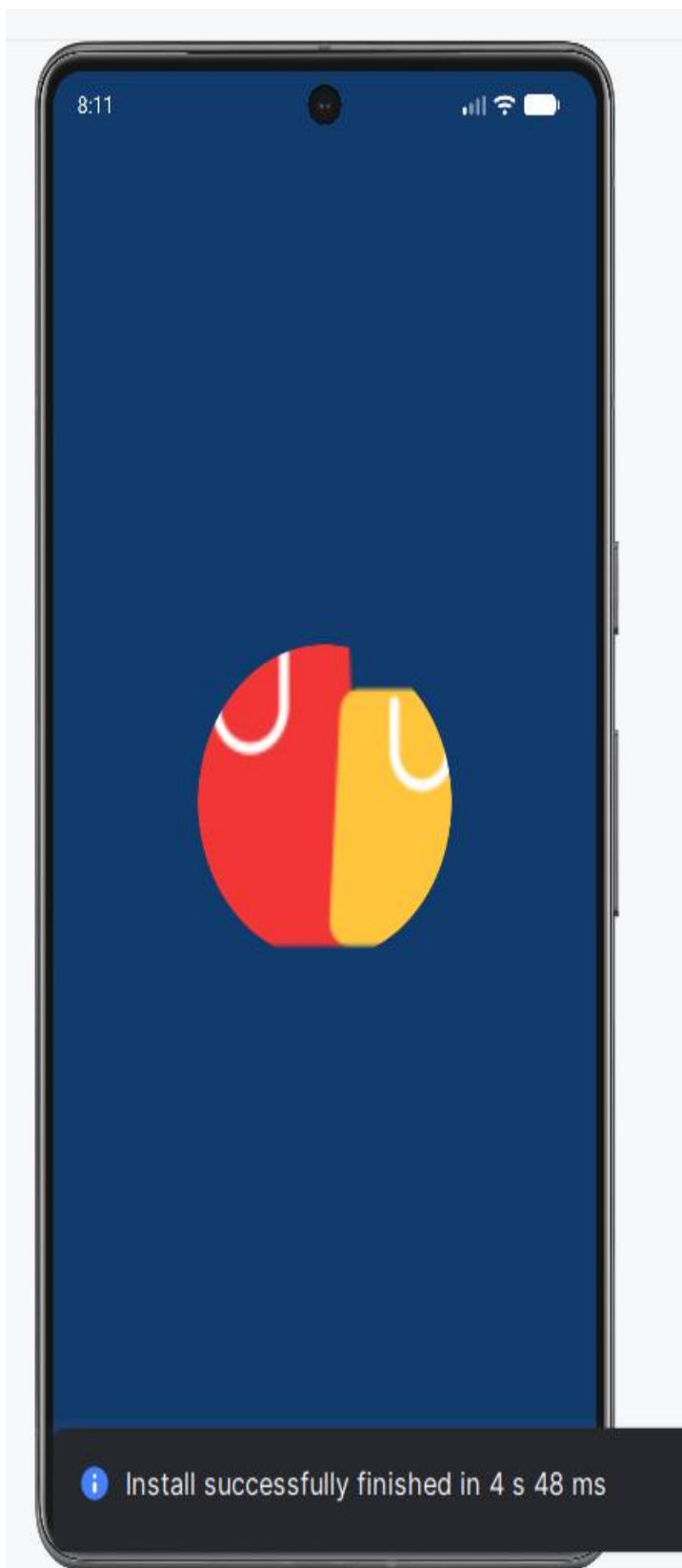
4. Firebase Configuration

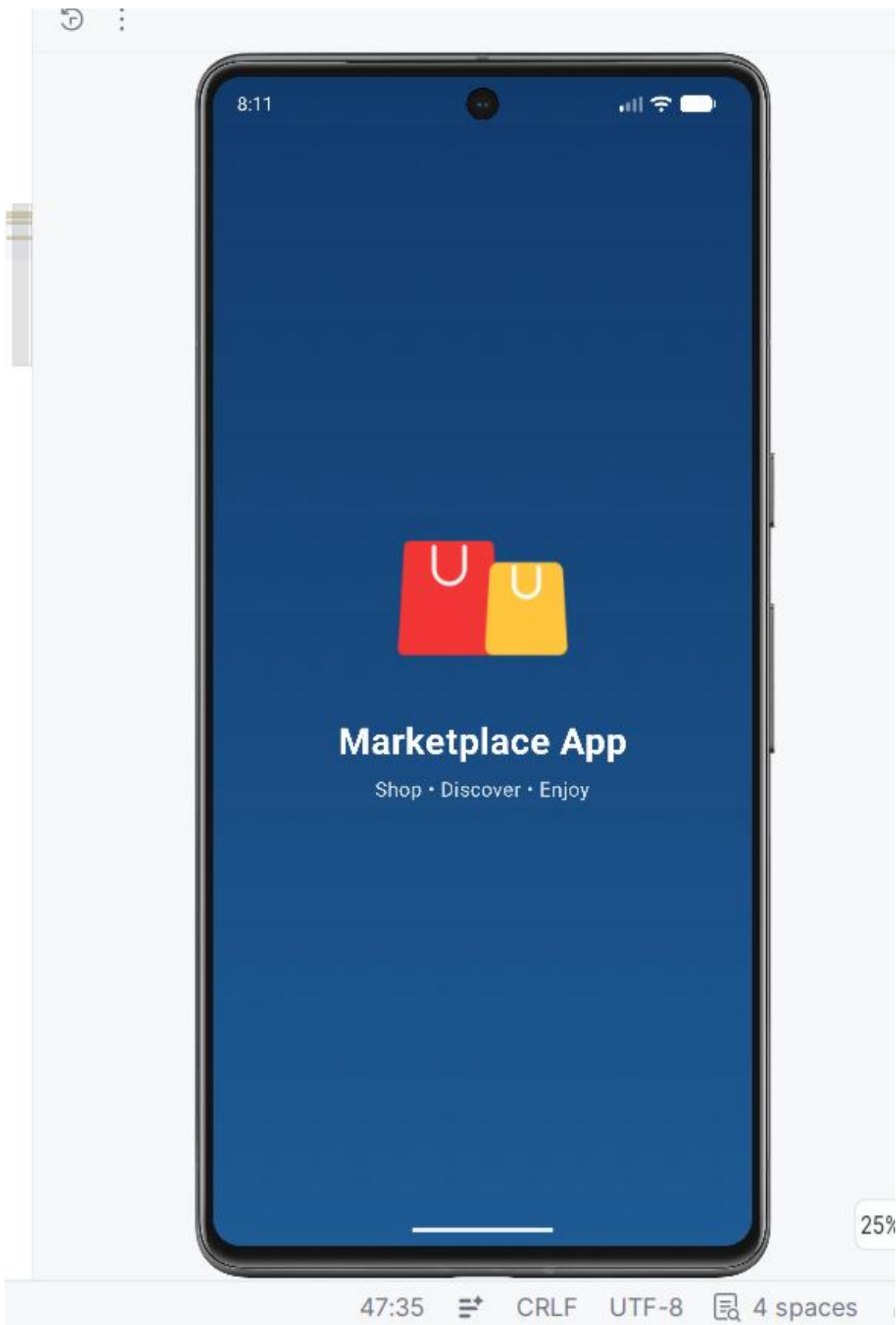
Firebase was connected to the application to provide authentication, Firestore database services, cloud functions and hosting.



5. Marketplace App Splash Screen

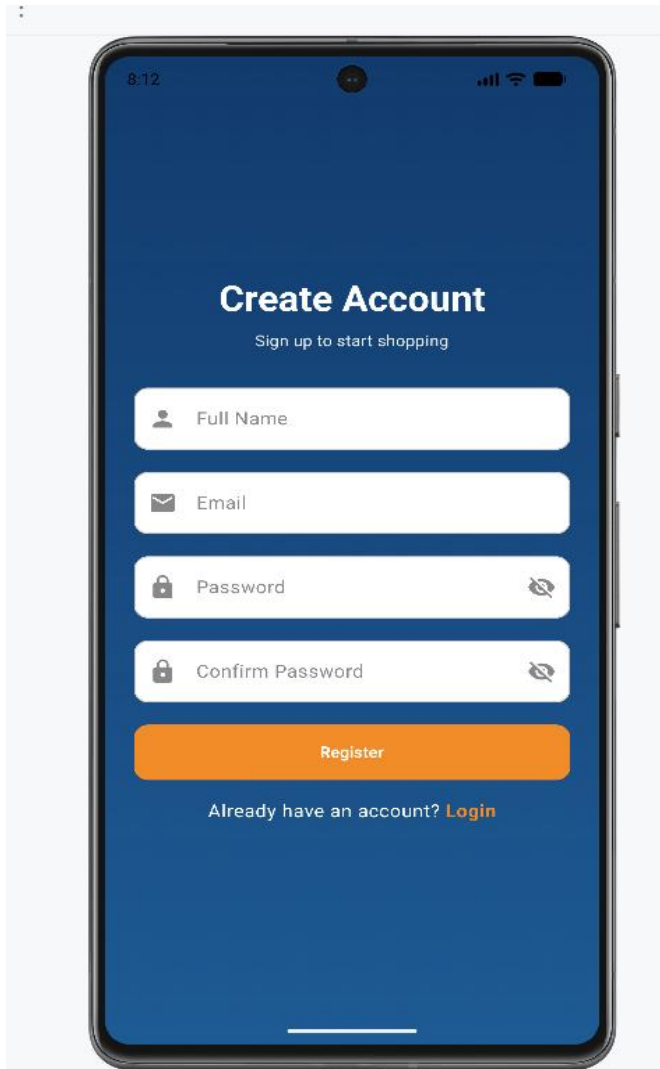
A branded splash screen was developed using the Marketplace App logo, application name and the application's blue colour scheme





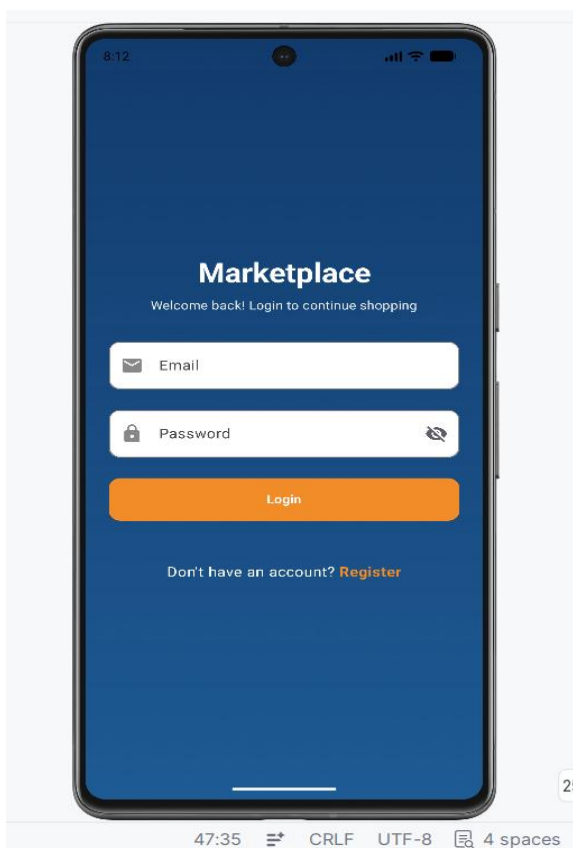
6. User Registration

A registration screen was implemented with full name, email, password and confirmation fields. Firebase Authentication creates the account and Firestore stores additional user information.



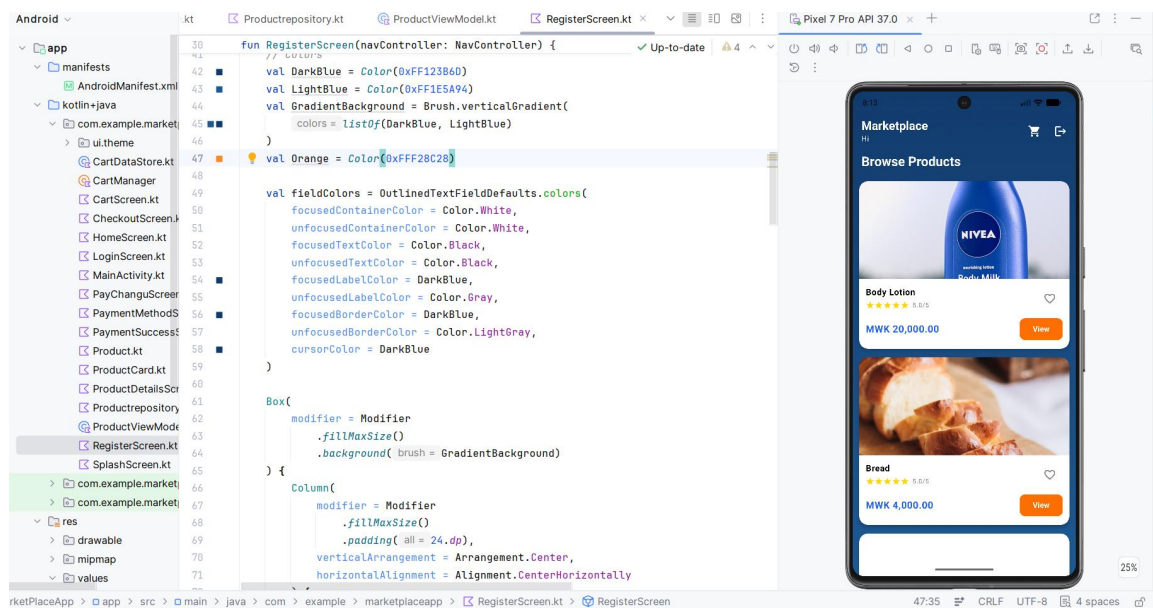
7. User Login

A login screen was implemented to authenticate registered users using Firebase Authentication.



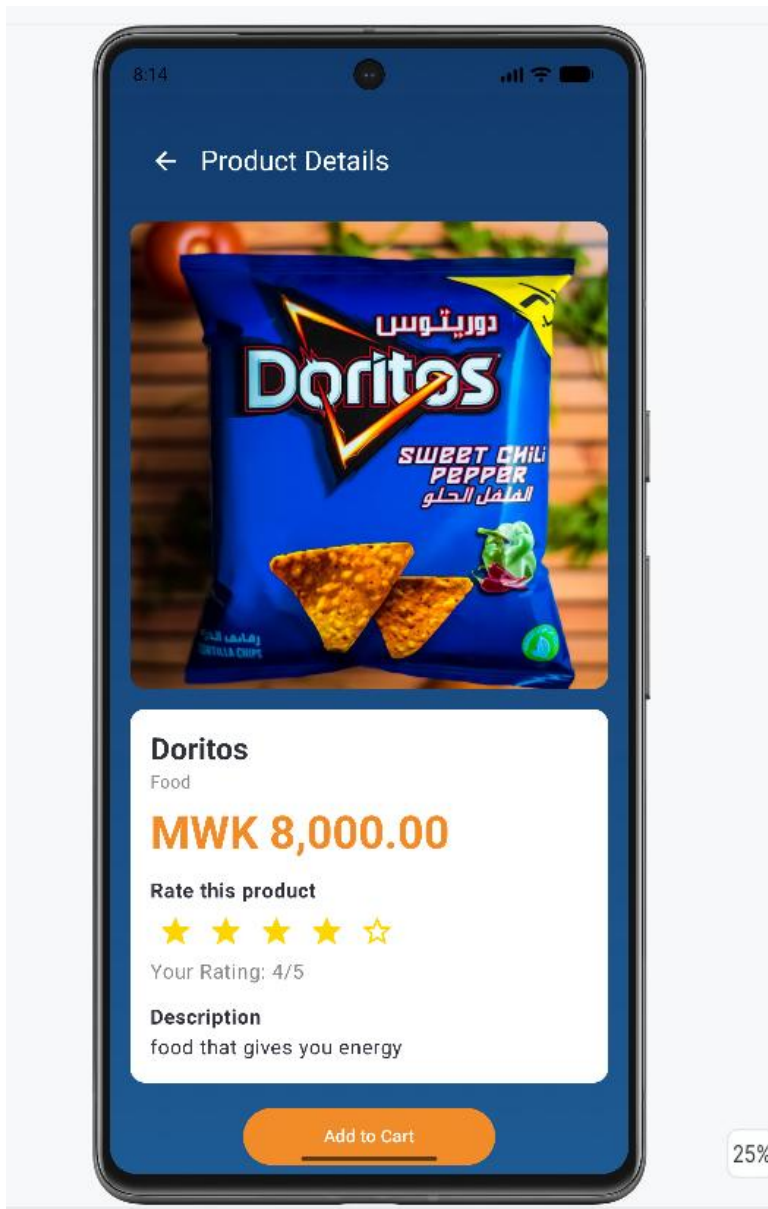
8. Home Screen and Product Catalogue

The home screen displays marketplace products and provides a user-friendly browsing experience.



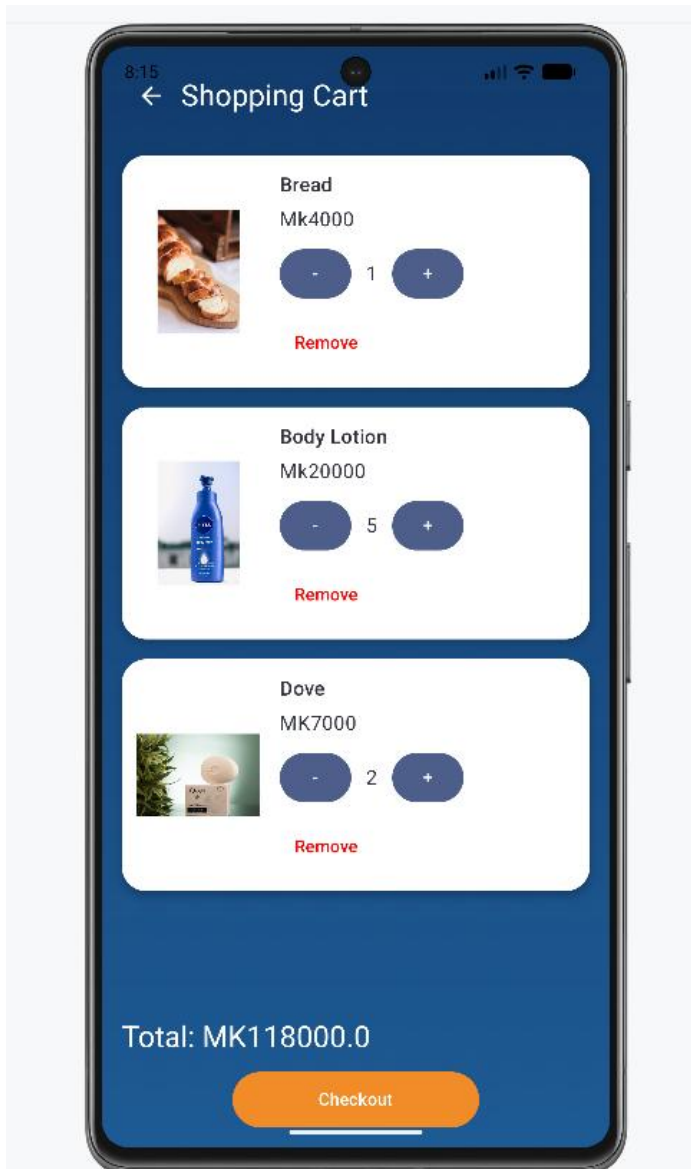
9. Product Details

The product details screen displays information about a selected product before it is added to the cart.



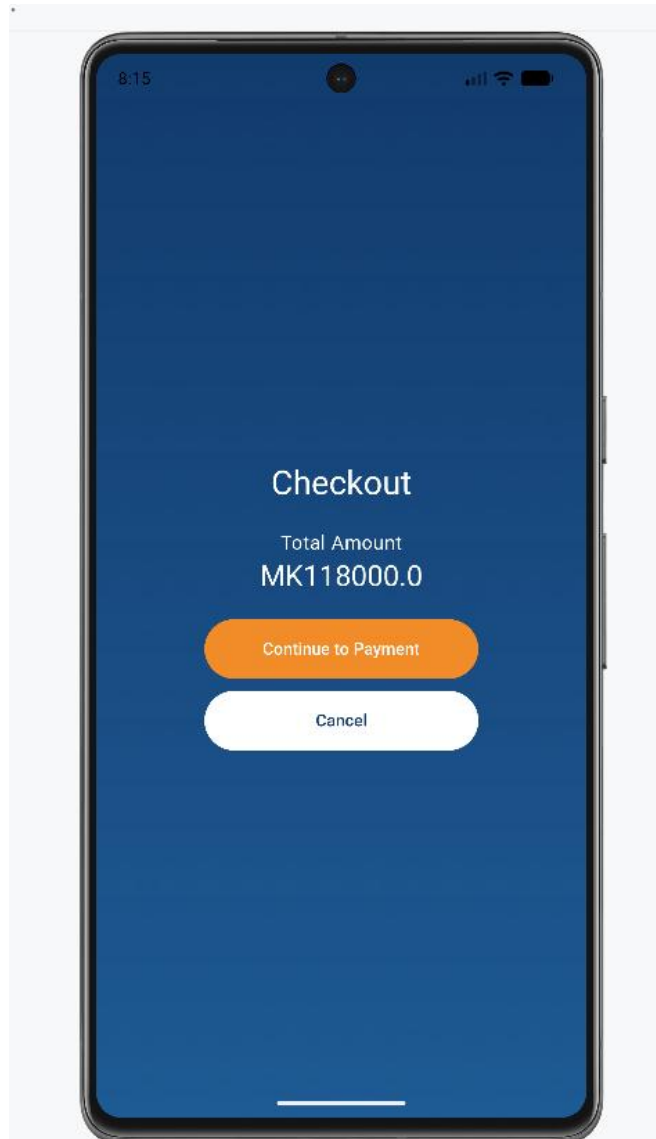
10. Shopping Cart

A shopping cart was implemented for managing selected products, quantities and the order total.



11. Checkout

The checkout screen allows users to review their order and total amount before selecting a payment method.



12. Payment Method

A payment method screen was created using the application's blue and orange colour scheme, with PayChangu as the payment option.

8:16



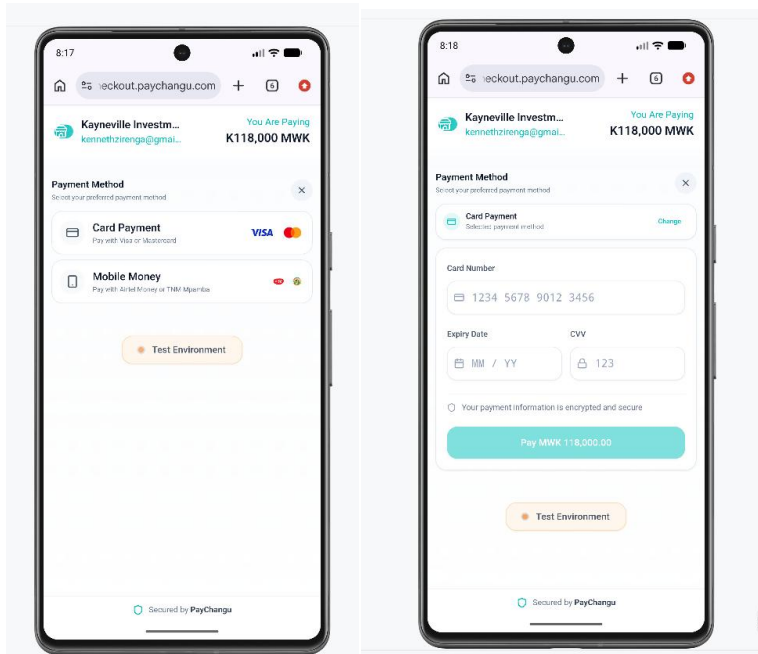
Choose Payment Method

Pay with Pay Changu

Back

13. PayChangu Integration

PayChangu was integrated through a Firebase Cloud Function. The application requests a checkout URL and opens the PayChangu payment page.



14. Payment Success

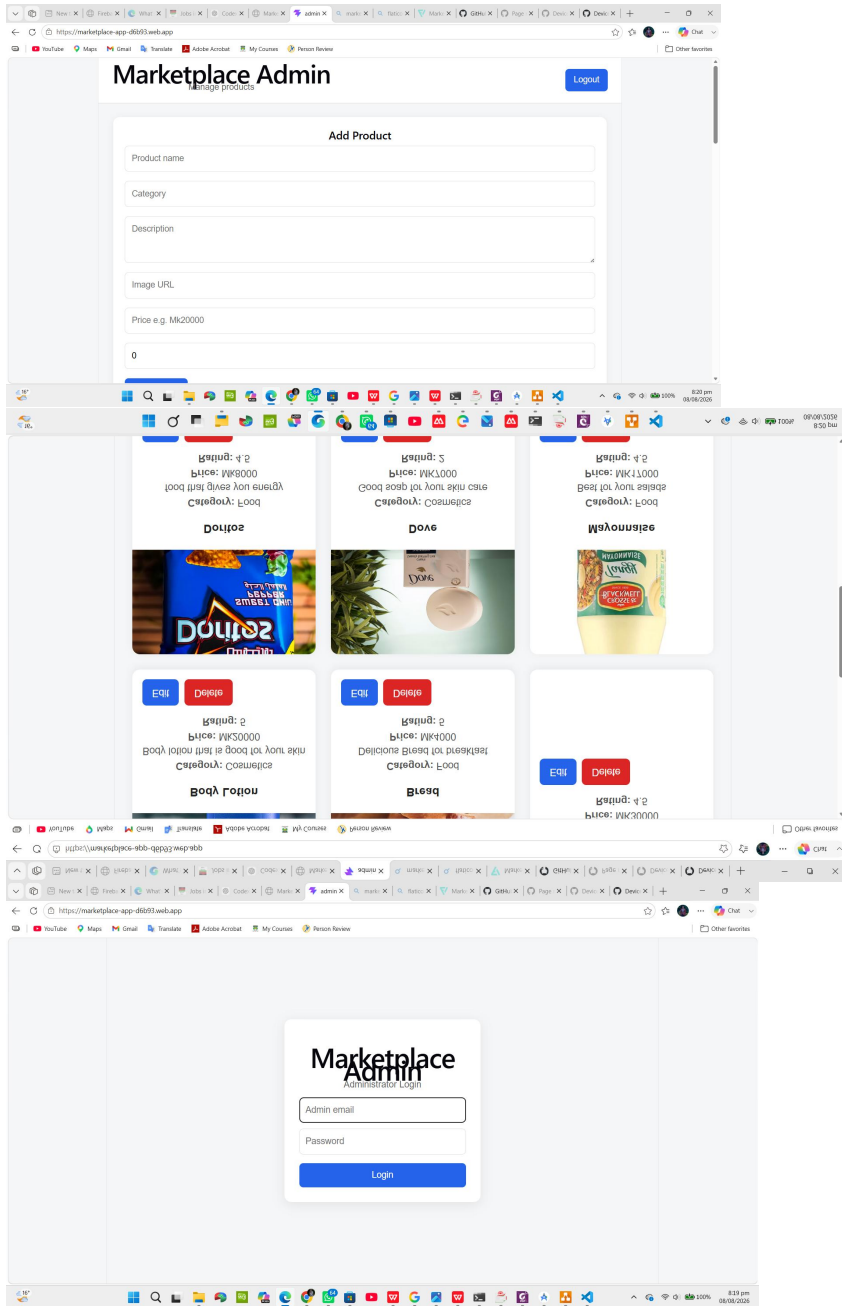
A payment return flow was implemented using a custom application URL scheme to recognize the payment-success return and display the payment success screen.

15. Firebase Cloud Functions

Firebase Cloud Functions handle server-side operations required by the application, including creation of the PayChangu payment request.

16. Administration Web Application

A separate React and Vite administration portal was developed for managing marketplace products through Firebase.



17. Firebase Hosting

The administration web application was built with Vite and deployed using Firebase Hosting.

18. Testing

The application was tested for registration, login, product browsing, cart, checkout, payment flow, Firebase operations and administration functions.

19. GitHub Repository

Complete project repository:

<https://github.com/kennethzirenga-cyber/MarketplaceApp>

20. Firebase Hosting

Administration portal:

<https://marketplace-app-d6b93.web.app>

21. Conclusion

The Marketplace App demonstrates a complete mobile marketplace solution using Kotlin, Jetpack Compose, Firebase, PayChangu and a web-based administration portal. The development process provided practical experience in mobile development, authentication, database services, payment integration, cloud functions, web administration, testing and GitHub version control.